Performance Evaluation of Masked Face Recognition using Deep Learning for Covid-19 Standard of Procedure (SOP) Compliance Monitoring

Mohd Norhisham Razali
Faculty of Business Management
Universiti Teknologi MARA Cawangan Sarawak, Kampus Samarahan
Kota Samarahan Sarawak, Malaysia
hishamrazali@uitm.edu.my

Ana Salwa Shafie
Faculty of Computer Science and Mathematic
Universiti Teknologi MARA Cawangan Pahang, Kampus Jengka, Pahang, Malaysia
anas674@uitm.edu.my

Rozita Hanapi
Faculty of Business Management
Universiti Teknologi MARA Cawangan Sarawak, Kampus Samarahan
Kota Samarahan Sarawak, Malaysia
rozita8282@uitm.edu.my

2021 6th IEEE International Conference on Recent Advances and Innovations in Engineering (ICRAIE) | 978-1-6654-3402-7/21/$31.00 ©2021 IEEE | DOI: 10.1109/ICRAIE52900.2021.9703986

***Abstract*— Wearing a face mask can reduce the risk of Covid-19 transmission. As a reason, creating an effective masked face recognition model is critical for the development of an autonomous face mask wearing monitoring system. Manual way of face mask wearing monitoring is a tedious task especially in the crowd and large public areas. Furthermore, masked face recognition is complex due to variety of face mask wearing image appearances such as occlusions, calibrations, scene complexity and the types of face mask used. This paper provides the performance evaluation of the Deep Convolutional Neural Network (CNN) model and machine learning classifiers for masked face recognition. Specifically, DENSENET201, NASNETLARGE, INCEPTIONRESNETV2 and EFFICIENTNET (EFFNET) as a feature extractor. Then, the extracted features are classified by using Support Vector Machine (SVM), Linear Support Vector Machine (LSVM), Decision Tree (DT), K-nearest neighbour (KNN) and Convolutional Neural Network (CNN). The recognition model is evaluated on the face mask detection dataset. The experiment results have shown that DENSENET201-SVM and EFFNET-LSVM obtained the best classification accuracy of 0.9972. However, EFFNET-LSVM has the advantage of better computational time of feature extraction, classification as well as the size of features.**

Keywords—Covid-19, face mask detection, deep learning, transfer learning

I. Introduction

The global catastrophic of the pandemic caused by Coronavirus (Covid-19) was not only a life threatening and paralyzing the public healthcare system, but also has severely affect the social and economic development in a country. As at 8 June 2021, there have been 173,331,478 Covid-19 confirmed cases reported to WHO [1]. There were millions of people are on the verge of becoming impoverished and 690 million of people would became undernourished [2]. The on-going mass vaccination program started in the early December 2020 can be seen as the hopes and solution to combat the killed virus. The effectiveness of the Covid-19 vaccine, on the other hand, is dependent on a number of circumstances [3] and yet, vaccination program will took years [4] in order to achieve at least 80% population vaccinated. Moreover, people are still may get infected and being infectious of the Covid-19 virus even after being vaccinated due to newly emerged virus variants [5]. Therefore, following the Covid-19 Standard of Procedure (SOP) is the best practice to follow in minimizing the risks of the Covid-19 spreads while waiting for herd immunity. Wearing face mask is the effective prevention measure SOP of Covid-19 transmissions [6]–[9]. However, the traditional way of monitoring the large group or crowd of people to ensure each individual are wearing the face mask correctly is quite challenging tasks. It necessitates more workers and resources, which not only exposes workers to the infection, but also prone to human mistake [8]. The Covid-19 SOP compliance from the public in a country are very important to ensure its effectiveness to prevent and to minimize risks of Covid-19 spreads. As reported in [10], the failure to adhere the Covid-19 SOP has caused to many new Covid-19 cases. Thus, efficient monitoring the public is crucial in ensuring all the Covid-19 SOP especially in wearing face mask properly. However, the current inspection to the large and crowd of people in public places such as in supermarkets, shopping mall, banks, school and any workplaces is not an easy task [4]. More resources, such as the hiring of more workers, will be necessary to accomplish these tasks. In addition to that, manual monitoring may lead to human mistake and also infeasible to locate many workers to keep in eye each individual to use and wear face mask properly [11].

The technology based on Machine learning and Computer Vision techniques can be used to develop a detection model. The model can be deployed in an automatic SOP compliance detection tool development to address the aforementioned problems of monitoring the large and crowd of people in public places. The current face mask wearing detection model does not cater for the detection of appropriate use of face masks and also trained in limited and artificial data [12]–[16]. Appropriate use of face mask is crucial to make sure it can work effectively in suppressing the Covid-19 transmissions [17]. Inappropriate use of masks including not to use a proper type of mask, wear mask under chin which expose nose and mouth and to leave nose or mouth uncovered. In addition to

978-1-6654-3402-7/21/$31.00 ©2021 IEEE

Authorized licensed use limited to: University of Canberra. Downloaded on November 10,2022 at 12:48:40 UTC from IEEE Xplore. Restrictions apply.

that, face mask recognition is challenging due to image occlusions, different calibrations, the scene complexity as well as the variety types of face mask in the market. Therefore, the objective of this paper is to analyze the applicability of CNNs in a masked face recognition model using face images. The analysis is conducted in feature extraction and classification aspects in creating a robust recognition model for masked face detection that can be deployed in real application.

The remainder of the paper is organized in the following way: Section II provide the previous studies conducted in face mask recognition. In Section III, the Methodology is described in more detail. Sections IV presents the experimental results and the outcomes as well as the evaluation of face mask recognition. Finally, conclusions and directions for the future studies are presented in Section V.

II. Related Works

The real-time and automatic monitoring of face of mask wearing Covid-19 SOP compliance are required in order to overcome the limitations of manual monitoring especially to deal with a large and crowd of people. Many previous research have applied deep learning approaches in face mask recognition [18]–[21].

The face mask detection model proposed by [22] hybridize the deep learning and classical machine learning classifier to be deployed in surveillance camera. The features are extracted based on ResNet50 transfer learning model and trained on SVM classifier. The accuracy of detection model yield on RMFD , SMFD and LFW datasets are 99.64%, 99.49% and 100% respectively. Another study conducted by [23] investigated the techniques to annotate and localize the face mask on MMD and FMD datasets. YOLOv2 and ResNet50 were used for the face detection and feature extraction respectively. While mean IoU is used for estimating the anchor boxes. The performance reported in this study was 81% of precision.

As wearing face mask correctly is important to ensure its effectiveness in suppressing Covid-19 transmissions, [12] proposed a face mask detection datasets for recognizing a proper use of face mask. The larger datasets are proposed in this study which are CMFD, IMFD and MaskedFace-Net. Initially, the face is detected by using Haar feature-based cascade classifiers. Next, the facial landmarks are detected based on specific keypoints facial landmarks detector. This is followed by mask to face mapping which create four classes correctly and incorrectly of wearing masks. However, the artificially created dataset might contained lot of noises as it is not based on the real data. Moreover, the data is limited to one type of face mask. Therefore, [13] use a real-world data which is Real-time medical masks dataset to proposed a face mask detection model based on SSDMNV2 using OpenCV Deep Neural Network (DNN), Tensorflow, Keras and MobileNetV2. The proposed model demonstrated 92.64% accuracy on Real-time medical masks dataset. However, the trained model is only based on single type of face mask and the images were not acquired based on surveillance camera which is commonly used in many public areas.

Meanwhile, the face mask detection model developed by [15] investigating the technique to locate the face in real time and determining the way of face mask is worn. SE-YOLOV3 is used to localize the face and the SE block with three steps; squeeze, excitation and scale to determine the correctness of wearing the face mask. The SE-YOLOV3 with SE-Darknet53 backbone has obtained the best performance with the Average Precision (AP) of 99.5% on PWFMD dataset.

Another study conducted by [24] proposed a real-time deep learning model on properly wearing of face mask using Keras RetinaNet object detection framework and combine with head thermal detector in order to measure the human temperature. The methods were evaluated on Medical-mask dataset with 81.31% accuracy on face mask detection and 99.7% accuracy for temperature detection. However, the data used for thermal head and facial images are less diverse.

III. Methodology

This section presents the steps undertaken to conduct this research which will be described in the following sections.

A. Experiments Preparation

The experiment in this study is performed by using Python libraries edited using Spyder 4.2.2 and tools. Specifically, the feature extractions and classifications are performed by using Scikit-learn and Keras libraries. Meanwhile, the operation system used is Windows 10 Pro with 8GB RAM, Intel Core i7 processor.

B. Masked Face Recognition Model Training

The masked face recognition model training consists of three main steps which are data preparation, feature extraction and classification model training.

1) Dataset

The performance of masked face recognition model is evaluated on face mask dataset [13]. The dataset is the collection of various public datasets comprising Kaggle Medical Mask dataset and Prajna Bhandary dataset available at PyImageSearch. The samples of the dataset is visualized in Figure 1.

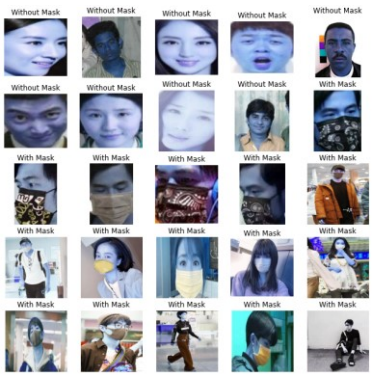

Fig. 1. Face Mask Dataset

Authorized licensed use limited to: University of Canberra. Downloaded on November 10,2022 at 12:48:40 UTC from IEEE Xplore. Restrictions apply.

The dataset contains 11042 images in JPEG format. It is divided into two categories: without mask and with mask, with 5521 photos in each category. The appearances and settings of images are diverse as it has different resolution size, captured in different lighting condition and camera viewpoints. Aside from that, the face mask comes in a variety of styles, types, and ways to wear it. The images are randomly divided into training and testing images for classification model training with 70:30 ratio.

2) Feature Extraction

The feature extraction of the images is performed using a transfer learning strategy, which employs a pre-trained model. The pre-trained model is trained using Convolutional Neural Network (CNN) on a huge dataset. Thus, the produced pre-trained model is generic enough to provide knowledge to make classification on certain objects. Furthermore, reusing the model may reduce the computational effort in training the large dataset as less parameters involved during the training due to the reduction number of layers. There are four recent pre-trained CNN model employed in this paper which are *DENSENET201*[25], *NASNETLARGE* [26], *INCEPTIONRESNETV2* [27] , and *EFFICIENTNET (EFFNET)* [28].

a) DENSENET201

The *DENSENET201* is proposed to maximize the information flow among the network's layers as all the current network topology use the short path between the sequence of layers. Hence, less parameters being used and as re-learning the redundant maps are not necessary. The information flow also can be improved, and the regularizing effect can deal with the overfitting problem on training the small dataset. *DENSENET201* architecture have demonstrated good performance on many recent applications. In the study conducted in [29], *DENSENET201* yield 96.22% accuracy on Covid-19 detection from chest X-Ray images. *DENSENET201* have also achieved the best performance in skin lesion classification [30], [31] with accuracy of 96.4% and 95%.

The total of trainable parameters is 20,013,928. The input of shape is *224 X 224 X 3* and reduce to *7 X 7 X 512* on the final network. The weights is based on Imagenet with avg_pool layer that means that global average pooling will be applied to the output of the last convolutional block, and thus the output of the model will be a 2-D tensor. Finally, 1920 attributes were produced for 1D classification and and (8,8,30) for 2-D classification.

b) NASNETLARGE

The base of *NASNETLARGE* model is Neural Architecture Search (NAS) framework that was applied on ImageNet dataset. The improvement of NAS framework have been made to reduce its computational effort by learning smaller dataset which is CIFAR-10 and transferred to trained architecture of ImageNet. To transfer the newly trained model, a NASNET search space is designed in order to ensure the complexity of the model is not dependent of the network's depth and image sizes. The searching mechanism is efficient and improved the generalization ability of NAS framework. Many previous studies have demonstrated great performance of *NASNETLARGE* model such as the rice diseases classification with accuracy of 85.9% [32], ECG signal classification for cardiac examination [33] with accuracy of 97.1%, lung nodule classification from CT lung images with accuracy of 88.28% [34] and skin lesion classification for nine categories of different diagnostic from dermoscopic images [35].

There are 88,753,150 trainable parameters for *NASNETLARGE* and take *331 X 331 X 3* as input size. To create features for both 1-D and 2-D classification, the prediction layer that created 1000 and (2, 2, 250) attributes was used.

c) INCEPTIONRESNETV2

The hybrid between Inception-v4 and ResNet model have been studied that produced *INCEPTIONRESNETV2*. The Inception-v4 is utilized with RESNET as it is trained deeply and may sustain the computational efficient. The integration between these models have simplified the complex architecture of Inception-v4 and also significantly improved the recognition performance. *INCEPTIONRESNETV2* model have been applied in many recognition problem in various domains. [36] evaluate various CNN models to differentiate between the types of pneumonia which are bacterial pneumonia, virus pneumonia and the normal lung from the chest x-ray images. The analysis revealed that the *INCEPTIONRESNETV2* yield top-3 recognition performance with accuracy of 92.4%. *INCEPTIONRESNETV2* is also have been explored in training the recognition model for diagnosing Covid-19 [38] other than the common pneumonia dieses from chest x-ray dataset and yield 97.1% accuracy [37].

The total trainable parameters for *INCEPTIONRESNETV2* is 55,813,192 and the input image size is 299 X 299 X 3. Then, the *avg_pool* is used that finally generate 1536 1-D features and (16,16,6) 2-D features.

d) EFFNET

The EFFNET introduced to tackle the problem in MobileNet and ShuffleNet which create too large network by making the network smaller and create a unified bottleneck. The unified bottleneck have decrease the computation time and optimize the dataflow. The performance evaluated on Cifar10, Street View House Numbers and German Traffic Sign Recognition Benchmark datasets which obtained promising accuracy and computational effort over MobileNet and ShuffleNet. The model developed by [39] for Tuberculosis detection from chest X-ray images have implemented EFFNET model and integrate Attention with Vision transformer as EFFNET is efficient and compact. Beside that, EFFNET is also effective for tracking the vehicle for traffic flow monitoring [40] with accuracy of 95.45% and food recognition system [41] to deal with diversity of indigenous foods with accuracy of 86.57%. Another study conducted by [42] have yield low degree of dispersion error (RMSE) by using the EFFNET features for counting animals to monitor the populations, specifically Stellar sea lions and African elephants.

Authorized licensed use limited to: University of Canberra. Downloaded on November 10,2022 at 12:48:40 UTC from IEEE Xplore. Restrictions apply.

In total, 5,288,548 trainable parameters in EFFNET and required *224 X 224 X 3* input image size that produce 1280 1-D features and (16,16,245) 2-D features.

3) Classification Model Training

The 1-D and 2-D features produced using *DENSENET201*, *NASNETLARGE*, *INCEPTIONRESNETV2*, and *EFFNET)* is evaluated by using various machine learning classifiers namely Support Vector Machine (SVM), Linear SVM (LSVM), Decision Tree (DT), k-nearest neighbor (KNN) and Convolutional Neural Network (CNN). The number of epoch, which refers to the number of times the model is trained, must be specified when using CNN for training. Thus, epoch number analysis ranging from 30,60,90, and 120 is carried out.

4) Performance Measurement

Accuracy was computed on the testing features to measure the classifiers' average generalisation performance. Over all samples, accuracy is defined as the number of samples correctly classified which is calculated using the formula presented in equation (1).

$$Accuracy = \frac{TP+TN}{(TP+FP)+(TN+FN)} \quad (1)$$

where TP represents the number of True Positive samples, TN represents the number of True Negative samples, FP represents the number of False Positive samples, and FN represents the number of False Negative samples from a confusion matrix.

Besides that, the evaluation is also conducted based on the computational time for feature extraction and classification as well as the features vector size.

IV. Experiments Results

This section describes the findings from the experiments. The findings comprise of performance evaluation of various CNN models for feature extraction and machine learning classifier, the computational time for classification and feature extraction as well as the feature length generation. Figure 2 shows the classification accuracy of the masked face recognition based on the features extracted using *DENSENET201, NASNETLARGE, INCEPTIONRESNETV2 AND EFFNET* and machine learning classifier using Support Vector Machine (SVM), Linear SVM (LSVM), Decision Tree (DT), k-nearest neighbor (KNN) and Convolutional Neural Network (CNN).

In overall, the results shown in Figure 2 demonstrated a good performance of all tested features and classifiers as all the classification accuracies are above 0.9. The highest accuracy was achieved by DENSENET201-SVM and EFFNET-LSVM, both of which were 0.9972. In the flipside, NASNETLARGE-KNN has lowest accuracy of 0.9218. Based on the graph pattern illustrated in Figure 1, the performance of all features are quite consistent towards all the classifiers except for NASNETLARGE.

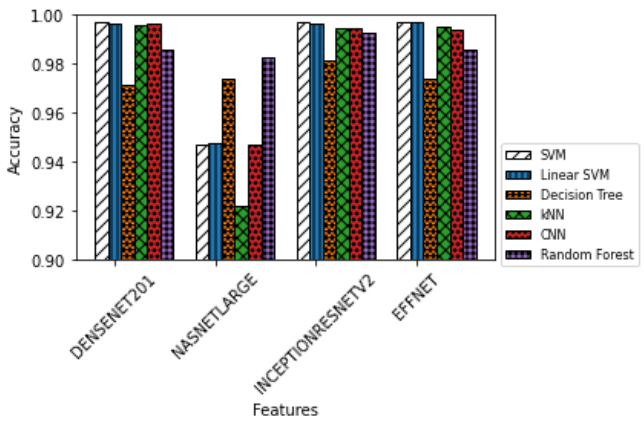

Fig. 2. Classification Accuracy based on Features

Figure 3 depicts the classification accuracy of all features grouped by the classifier in order to evaluate the classifiers' efficacy with respect to the features.

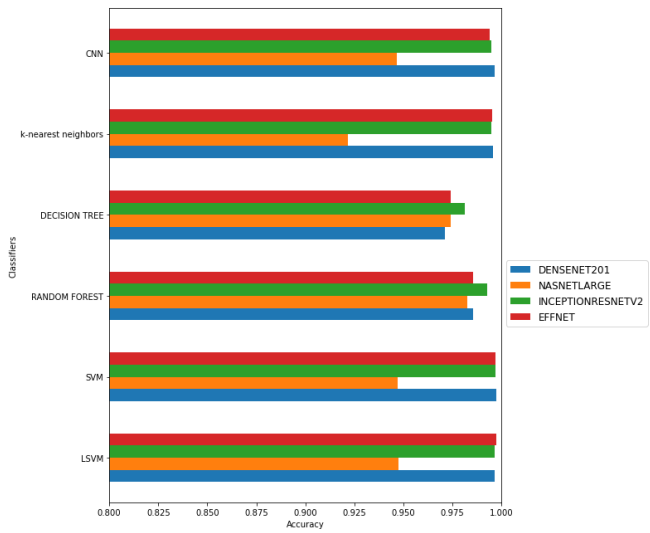

Fig. 3. Classification Accuracy based on Classifiers

On CNN, KNN, SVM, and LSVM, a similar pattern of performance can be seen, where it performs excellently with all features except NASNETLARGE. However, DT and RF performance towards all the features are quite consistent despite of the accuracy is not high as the other classifiers.

Figure 4 shows the analysis CNN performance using different epoch.

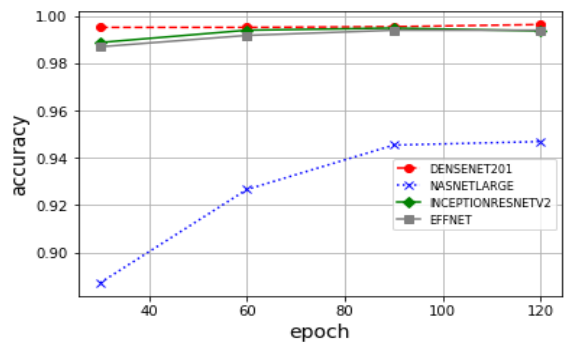

Fig. 4. Classification Accuracy of various features using CNN

Authorized licensed use limited to: University of Canberra. Downloaded on November 10,2022 at 12:48:40 UTC from IEEE Xplore. Restrictions apply.

The graph in Figure 4 shows that the *DENSENET201*, *INCEPTIONV2* AND *EFFNET* were able to obtain good accuracy with a minimal number of epoch (30). However, increasing the number of epoch to 60, 90, and 120 does not improve their accuracy substantially. *NASNETLARGE*, on the other hand, requires at least 60 epoch to attain 0.9266 accuracy. The highest classification accuracy of *DENSENET201* and *NASNETLARGE* was recorded using epoch 120, which was 0.9964 and 0.9469, respectively, as shown in Table III. *INCEPTIONRESNETV2* and *EFFNET* with epoch 90, on the other hand, used to have the best classification accuracy. The number of epoch plays important role in determining the computation complexity as larger epoch will prolong the training and testing duration. The graph shown in Figure 5, Figure 6, Figure 7 and Figure 8 were the training and testing of CNN of all the features over the number of epoch.

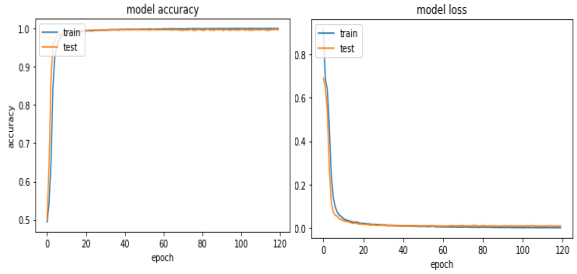

Fig. 5. Classification Performance of DENSENET201 using CNN

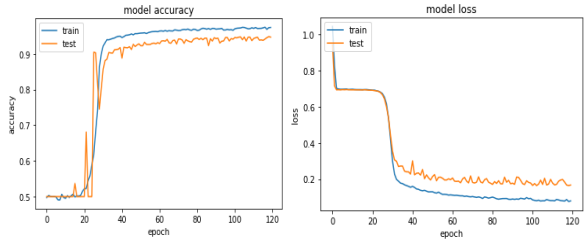

Fig. 6. Classification Performance of NASNETLARGE using CNN

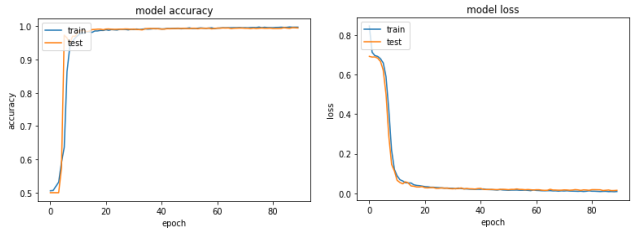

Fig. 7. Classification Performance of INCEPTION-RESNETV2 using CNN

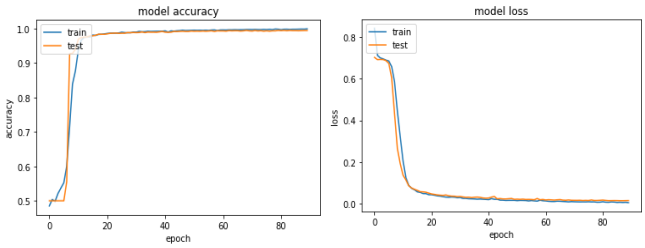

Fig. 8. Classification Performance of EFFNET using CNN

The graph shown Figure 5, Figure 7 and Figure 8 shown minimal gap between training and testing in the model loss that indicate a good classification performance with the selected epoch. However, larger gap between training and testing in the model lost was demonstrated using *NASNETLARGE* feature which probably require much larger epoch to narrow down the gap size.

Figure 9 shows the classification computation time or training time for each feature and classifier.

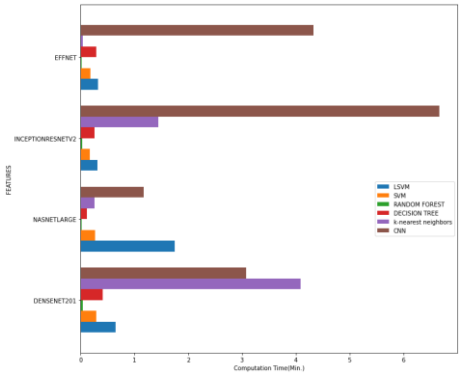

Fig. 9. Classification Computation Time (Minutes)

Based on the graph in Figure 9, the training duration for CNN is quite high on all features with average of 3.8126 minutes followed by KNN with average of 1.4569 minutes. The training time for *INCEPTIONRESNETV2-CNN* recorded the highest computation time with 6.6664 minutes. Random Forest perform the faster with average of 0.0297 minutes followed by SVM (0.22882 minutes), DT (0.270075 minutes) and LSVM (0.759225 minutes). The training duration on *NASNETLARGE* is the smallest with average of 0.5962 minutes followed by EFFNET (0.8662 minutes), *DENSENET* (1.4272 minutes) and Inceptionresnetv2 1.4817 minutes.

Figure 10 presents the computation time for feature extraction and Figure shows the features length generated by each feature.

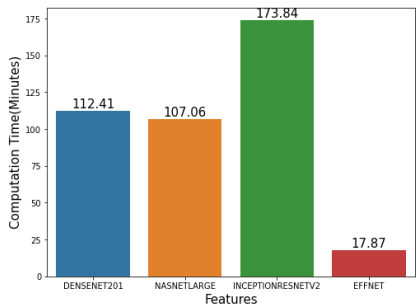

Fig. 10. Feature Extraction Computation Time

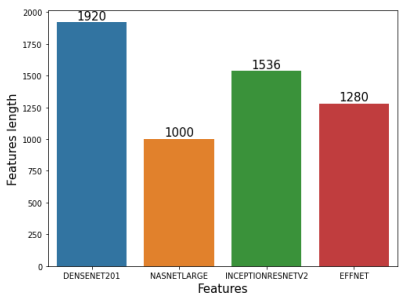

Fig. 11. Features Length

Authorized licensed use limited to: University of Canberra. Downloaded on November 10,2022 at 12:48:40 UTC from IEEE Xplore. Restrictions apply.

As shown in Figure 10, *INCEPTIONRESNETV2* took longest duration to extract the features even though *NASNETLARGE* have more parameters compared to Inceptionresnetv2 and the size of features generated is lesser than DENSENET201 as can be seen in Figure 10. Meanwhile, the extraction computation time for *EFFNET* is much far lower compared to the other features as it has much smaller number of parameters. The *NASNETLARGE* generates the smallest size of features which probably caused the classification computation time can be performed faster as described in Figure 9.

In summary from these analysis, *DENSENET201-SVM* and *EFFNET-LSVM* have obtained the best classification accuracy of 0.9972. However, when other aspects such as computational time for classification and feature extraction, the length of generated features, and the number of training parameters are considered, *EFFNET-LSVM* appears as the best. In particular, the feature extraction computational time taken by *DENSENET201* are more than 6 times higher than EFFNET, the size of features generated by *DENSENET201* are about 30% larger than *EFFNET* and the training parameters used by *DENSENET201* is four times bigger than *EFFNET*.

V. Conclusion and Future Works

The Covid-19 pandemic has severely affect the way of human live nowadays. Despite a person have been fully vaccinated, but they are still susceptible to Covid-19 virus which still can carry and transmit the virus to other people. Therefore, the new norms such as wearing mask are still need to be always practiced. This paper has conducted the comprehensive analysis of recognition model performance for masked face using CNN and machine learning models. The recognition model is crucial for the development an automated Covid-19 Standard of Procedure (SOP) compliance monitoring. In overall, the CNN models by using *DENSENET201*, *NASNETLARGE*, *INCEPTIONRESNETV2*, and *EFFICIENTNET (EFFNET)* are very effective for masked face recognition. However, the EFFNET model with LSVM classifier get the best performance in term of accuracy, feature extraction and classification computation time as well as the size of features. Future work should evaluate using more datasets, acquire the local or real dataset and deploy the recognition model into real application.

References

[1] WHO, "WHO Coronavirus (COVID-19) Dashboard," 2021. https://covid19.who.int/ (accessed Jun. 09, 2021).

[2] K. Chriscaden, "Impact of COVID-19 on people's livelihoods, their health and our food systems," 2021. https://www.who.int/news/item/13-10-2020-impact-of-covid-19-on-people's-livelihoods-their-health-and-our-food-systems (accessed Jun. 09, 2021).

[3] WHO, "Coronavirus disease (COVID-19): Vaccines," 2020. https://www.who.int/news-room/q-a-detail/coronavirus-disease-(covid-19)-vaccines?topicsurvey=v8kj13)&gclid=Cj0KCQjwzYGGBhCTARIsAHdMTQxWk2kv9D843zx5BsFidTIvfGtLoa0psFLlrdG33l-y5JlenPyfxWkaAkASEALw_wcB.

[4] K. Suganeswaran, K. Dhileephan, N. Nithyavathy, P. Ganeshan, S. Arunkumar, and A. J. Antony, "Machine Learning based Mechanism for Crowd Mobilization and Control," *Proc. 6th Int. Conf. Inven. Comput. Technol. ICICT 2021*, pp. 1334–1339, 2021, doi: 10.1109/ICICT50816.2021.9358631.

[5] T. S. Times, "40 healthcare workers in Malaysia infected after Covid-19 vaccination reported less severe symptomse," *The Straits Times*, 2021. https://www.straitstimes.com/asia/se-asia/healthcare-workers-in-malaysia-infected-after-covid-19-vaccination-reported-less-severe (accessed Jun. 09, 2021).

[6] S. Suryadi, E. Kurniawan, H. Adinanta, B. H. Sirenden, J. A. Prakosa, and P. Purwowibowo, "On the Comparison of Social Distancing Violation Detectors with Graphical Processing Unit Support," *Proceeding - 2020 Int. Conf. Radar, Antenna, Microwave, Electron. Telecommun. ICRAMET 2020*, pp. 337–342, 2020, doi: 10.1109/ICRAMET51080.2020.9298574.

[7] J. Shah, M. Chandaliya, H. Bhuta, and P. Kanani, "Social Distancing Detection Using Computer Vision," pp. 1359–1365, 2021, doi: 10.1109/iccmc51019.2021.9418312.

[8] S. Srinivasan, R. Rujula Singh, R. R. Biradar, and S. A. Revathi, "COVID-19 monitoring system using social distancing and face mask detection on surveillance video datasets," *2021 Int. Conf. Emerg. Smart Comput. Informatics, ESCI 2021*, pp. 449–455, 2021, doi: 10.1109/ESCI50559.2021.9396783.

[9] T. Ikram, A. Saeed, N. U. Ayn, M. A. Tahir, and R. Mumtaz, "A review of the prevalent ICT techniques used for COVID-19 SOP violation detection," *HONET 2020 - IEEE 17th Int. Conf. Smart Communities Improv. Qual. Life using ICT, IoT AI*, pp. 194–198, 2020, doi: 10.1109/HONET50430.2020.9322821.

[10] The Star, "Health DG: Rise in Covid-19 cases due to complacency, non-compliance with SOP," *The Star Online*, Jul. 23, 2020.

[11] P. Somaldo, F. A. Ferdiansyah, G. Jati, and W. Jatmiko, "Developing Smart COVID-19 Social Distancing Surveillance Drone using YOLO Implemented in Robot Operating System simulation environment," *IEEE Reg. 10 Humanit. Technol. Conf. R10-HTC*, vol. 2020-Decem, 2020, doi: 10.1109/R10-HTC49770.2020.9357040.

[12] A. Cabani, K. Hammoudi, H. Benhabiles, and M. Melkemi, "MaskedFace-Net – A dataset of correctly/incorrectly masked face images in the context of COVID-19," *Smart Heal.*, vol. 19, no. November 2020, p. 100144, 2021, doi: 10.1016/j.smhl.2020.100144.

[13] P. Nagrath, R. Jain, A. Madan, R. Arora, P. Kataria, and J. Hemanth, "SSDMNV2: A real time DNN-based face mask detection system using single shot multibox detector and MobileNetV2," *Sustain. Cities Soc.*, vol. 66, no. August 2020, p. 102692, 2021, doi: 10.1016/j.scs.2020.102692.

[14] K. Suresh, M. B. Palangappa, and S. Bhuvan, "Face Mask Detection by using Optimistic Convolutional Neural Network," *Proc. 6th Int. Conf. Inven. Comput. Technol. ICICT 2021*, pp. 1084–1089, 2021, doi: 10.1109/ICICT50816.2021.9358653.

[15] X. Jiang, T. Gao, Z. Zhu, and Y. Zhao, "Real-time face mask detection method based on yolov3," *Electron.*, vol. 10, no. 7, pp. 1–17, 2021, doi: 10.3390/electronics10070837.

[16] N. A. A. Saleh, H. M. Ertunc, R. A. A. Saleh, and M. A. Rassam, "A Simple Mask Detection Model Based on A Multi-Layer Perception Neural Network," *2021 Int. Conf. Technol. Sci. Adm. ICTSA 2021*, pp. 0–4, 2021, doi: 10.1109/ICTSA52017.2021.9406523.

[17] WHO, "All about masks in the context of COVID-19," 2020. https://www.who.int/emergencies/diseases/novel-coronavirus-2019/advice-for-public/when-and-how-to-use-masks?gclid=Cj0KCQjwzYGGBhCTARIsAHdMTQzCjoMebbzTBQAvjOsP_lWihKv74G6SHDHzO4OtEJnby3jeP0CrrAkaAjD4EALw_wcB (accessed Jun. 10, 2021).

[18] H. Adusumalli, D. Kalyani, R. K. Sri, M. Pratapteja, and P. V. R. D. P. Rao, "Face Mask Detection Using OpenCV," *Proc. 3rd Int. Conf. Intell. Commun. Technol. Virtual Mob. Networks, ICICV 2021*, no. Icicv, pp. 1304–1309, 2021, doi: 10.1109/ICICV50876.2021.9388375.

[19] B. Lin, Y. Mu, Z. Fu, C. Li, and X. Duan, "A Network for Detecting Facial Features during the COVID-19 Epidemic," *ACM Int. Conf. Proceeding Ser.*, pp. 141–146, 2021, doi: 10.1145/3448218.3448240.

[20] M. Yu, S. Zou, A. Jia, and X. Cheng, "Recognition of the standardization of wearing masks during the epidemic of COVID-19," pp. 728–732, 2021, doi: 10.1109/ipec51340.2021.9421236.

[21] S. Degadwala, D. Vyas, U. Chakraborty, A. R. Dider, and H. Biswas, "Yolo-v4 Deep Learning Model for Medical Face Mask Detection," *Proc. - Int. Conf. Artif. Intell. Smart Syst. ICAIS 2021*, pp. 209–213, 2021, doi: 10.1109/ICAIS50930.2021.9395857.

[22] M. Loey, G. Manogaran, M. H. N. Taha, and N. E. M. Khalifa, "A hybrid deep transfer learning model with machine learning methods for face mask detection in the era of the COVID-19 pandemic," *Meas. J.*

Authorized licensed use limited to: University of Canberra. Downloaded on November 10,2022 at 12:48:40 UTC from IEEE Xplore. Restrictions apply.

Int. Meas. Confed., vol. 167, no. July 2020, p. 108288, 2021, doi: 10.1016/j.measurement.2020.108288.

[23] M. Loey, G. Manogaran, M. H. N. Taha, and N. E. M. Khalifa, "Fighting against COVID-19: A novel deep learning model based on YOLO-v2 with ResNet-50 for medical face mask detection," *Sustain. Cities Soc.*, vol. 65, no. November 2020, p. 102600, 2021, doi: 10.1016/j.scs.2020.102600.

[24] I. Farady, C. Y. Lin, A. Rojanasarit, K. Prompol, and F. Akhyar, "Mask Classification and Head Temperature Detection Combined with Deep Learning Networks," *2020 2nd Int. Conf. Broadband Commun. Wirel. Sensors Powering, BCWSP 2020*, pp. 74–78, 2020, doi: 10.1109/BCWSP50066.2020.9249454.

[25] G. Huang, Z. Liu, L. Van Der Maaten, and K. Q. Weinberger, "Densely connected convolutional networks," *Proc. - 30th IEEE Conf. Comput. Vis. Pattern Recognition, CVPR 2017*, vol. 2017-January, pp. 2261–2269, 2017, doi: 10.1109/CVPR.2017.243.

[26] B. Zoph, V. Vasudevan, J. Shlens, and Q. V. Le, "Learning Transferable Architectures for Scalable Image Recognition," *Proc. IEEE Comput. Soc. Conf. Comput. Vis. Pattern Recognit.*, pp. 8697–8710, 2018, doi: 10.1109/CVPR.2018.00907.

[27] C. Szegedy, S. Ioffe, V. Vanhoucke, and A. A. Alemi, "Inception-v4, inception-ResNet and the impact of residual connections on learning," *31st AAAI Conf. Artif. Intell. AAAI 2017*, pp. 4278–4284, 2017.

[28] I. Freeman, L. Roese-Koerner, and A. Kummert, "Effnet: An Efficient Structure for Convolutional Neural Networks," *Proc. - Int. Conf. Image Process. ICIP*, pp. 6–10, 2018, doi: 10.1109/ICIP.2018.8451339.

[29] M. M. R. Khan *et al.*, "Automatic Detection of COVID-19 Disease in Chest X-Ray Images using Deep Neural Networks," *IEEE Reg. 10 Humanit. Technol. Conf. R10-HTC*, vol. 2020-December, pp. 7–12, 2020, doi: 10.1109/R10-HTC49770.2020.9357034.

[30] C. Zhao, R. Shuai, L. Ma, W. Liu, Di. Hu, and M. Wu, "Dermoscopy Image Classification Based on StyleGAN and DenseNet201," *IEEE Access*, vol. 9, pp. 8659–8679, 2021, doi: 10.1109/ACCESS.2021.3049600.

[31] S. P. Godlin Jasil and V. Ulagamuthalvi, "Skin lesion classification using pre-trained DenseNet201 deep neural network," *2021 3rd Int. Conf. Signal Process. Commun. ICPSC 2021*, no. May, pp. 393–396, 2021, doi: 10.1109/ICSPC51351.2021.9451818.

[32] V. K. Shrivastava, M. K. Pradhan, and M. P. Thakur, "Neural Networks for Rice Plant Disease Classification," pp. 1023–1030, 2021.

[33] C. Cordoș, L. Mihailă, P. Faragó, and S. Hintea, "ECG signal classification using Convolutional Neural Networks for Biometric Identification," pp. 167–170, 2021.

[34] R. V. M. Da Nóbrega, S. A. Peixoto, S. P. P. Da Silva, and P. P. R. Filho, "Lung Nodule Classification via Deep Transfer Learning in CT Lung Images," *Proc. - IEEE Symp. Comput. Med. Syst.*, vol. 2018-June, pp. 244–249, 2018, doi: 10.1109/CBMS.2018.00050.

[35] S. A. A. Ahmed, B. Yanikoglu, O. Goksu, and E. Aptoula, "Skin Lesion Classification with Deep CNN Ensembles," *2020 28th Signal Process. Commun. Appl. Conf. SIU 2020 - Proc.*, pp. 1–4, 2020, doi: 10.1109/SIU49456.2020.9302125.

[36] A. Manickam, J. Jiang, Y. Zhou, A. Sagar, R. Soundrapandiyan, and R. Dinesh Jackson Samuel, "Automated pneumonia detection on chest X-ray images: A deep learning approach with different optimizers and transfer learning architectures," *Meas. J. Int. Meas. Confed.*, vol. 184, no. July, p. 109953, 2021, doi: 10.1016/j.measurement.2021.109953.

[37] A. K. Rangarajan and H. K. Ramachandran, "A preliminary analysis of AI based smartphone application for diagnosis of COVID-19 using chest X-ray images," *Expert Syst. Appl.*, vol. 183, no. January, p. 115401, 2021, doi: 10.1016/j.eswa.2021.115401.

[38] Y. Cho, S. H. Hwang, Y. W. Oh, B. J. Ham, M. J. Kim, and B. J. Park, "Deep convolution neural networks to differentiate between COVID-19 and other pulmonary abnormalities on chest radiographs: Evaluation using internal and external datasets," *Int. J. Imaging Syst. Technol.*, vol. 31, no. 3, pp. 1087–1104, 2021, doi: 10.1002/ima.22595.

[39] L. T. Duong, N. H. Le, T. B. Tran, V. M. Ngo, and P. T. Nguyen, "Detection of tuberculosis from chest X-ray images: Boosting the performance with vision transformer and transfer learning," *Expert Syst. Appl.*, vol. 184, no. January, p. 115519, 2021, doi: 10.1016/j.eswa.2021.115519.

[40] J. Azimjonov and A. Özmen, "A real-time vehicle detection and a novel vehicle tracking systems for estimating and monitoring traffic flow on highways," *Adv. Eng. Informatics*, vol. 50, no. March, p. 101393, 2021, doi: 10.1016/j.aei.2021.101393.

[41] M. N. Razali *et al.*, "Indigenous food recognition model based on various convolutional neural network architectures for gastronomic tourism business analytics," *Inf.*, vol. 12, no. 8, 2021, doi: 10.3390/info12080322.

[42] C. Padubidri, A. Kamilaris, S. Karatsiolis, and J. Kamminga, "Counting sea lions and elephants from aerial photography using deep learning with density maps," *Anim. Biotelemetry*, vol. 9, no. 1, pp. 1–10, 2021, doi: 10.1186/s40317-021-00247-x.

Authorized licensed use limited to: University of Canberra. Downloaded on November 10,2022 at 12:48:40 UTC from IEEE Xplore. Restrictions apply.